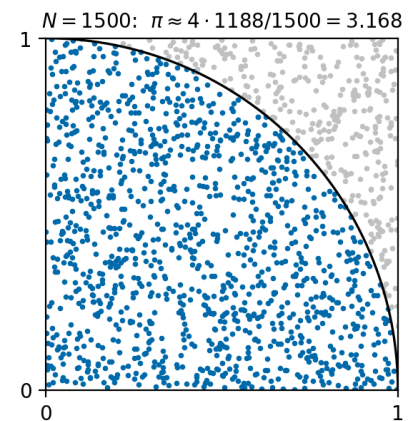


First Steps with Agentic AI

In this course you may use AI tools. *Agentic* tools such as Claude Code do not only answer questions – they read and write files, run commands and iterate on their own.

Today you get a first impression of how they work, where they help, and why you still have to understand and check every result.

The agent writes the code – you verify it.



1 Setup.

- Install Python with `numpy`, `scipy`, `matplotlib`, and `git`. Check with `python -c "import numpy, scipy, matplotlib"`.
- Install an agentic coding tool (e.g. Claude Code; alternatives: Codex CLI, Gemini CLI, GitHub Copilot agent mode). Create an empty folder, run `git init`, start the agent there and ask it what it can do.
Which actions does the agent ask permission for, and why?

2 A first task: estimate π by Monte Carlo. Draw N random points in the unit square; the fraction inside the quarter circle approximates $\pi/4$ (see figure).

- Ask the agent to write a script that estimates π for $N = 10^2 \dots 10^7$ and plots the error versus N . Follow what it does (files, commands, output).
Which steps did the agent take without being asked?
- Read the code and run it yourself. Commit the result with `git commit`.
How does the error scale with N , and why?
Do you trust the plot?
How did you check it?
- Ask for a change, e.g. a vectorized version and a timing comparison with a Python loop. Review the changes with `git diff` before accepting them.
Which version is faster and by how much?
Did the agent change more than you asked for?
- Ask the agent to make the method converge faster than $1/\sqrt{N}$.
What does it propose?
Is its claim true? Verify it numerically.
- Create a short document which summarizes your results.

3 Reflection.

- Where did the agent save you time, and where did you have to correct or verify it?
- What should you write into the prompt (or a project file such as `CLAUDE.md`) so that the agent works the way you want?
- In the following exercises you must be able to explain every line of your code. How do you make sure of that when an agent wrote it?